BAIRE CATEGORY THEOREM FOR A COMPLETE LINEARLY TOPOLOGIZED ABELIAN GROUP WITH A COUNTABLE FUNDAMENTAL SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Linearly topologized abelian group with a countable fundamental system of neighbourhoods of zero. definition discussion of the same section, source lines 8996-9022, before the counted statement. Supplies the nested neighbourhood basis M_n and the meaning of completeness used to take the limit.
- Completeness of a topological abelian group in terms of Cauchy sequences. used at proof lines 9067-9069 to produce the limit y of the constructed sequence. Turns the constructed sequence, whose successive differences lie in the shrinking neighbourhoods, into an actual element y of M.
- A subset is dense if and only if it meets every nonempty open subset. used at proof lines 9057-9058 and 9061-9062 when the y_i are chosen inside the open cosets x + M_k and y_1 + M_{k_1}. Provides the points y_i in U_i at every step of the induction.

Source reference. Original lemma 0CQU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15974–15974. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2 (Baire category theorem)**.** *Let M be a topological abelian group. Assume M is linearly topologized, complete, and has a countable fundamental system of neighbourhoods of 0. If $U_n \subset M$, $n \geq 1$ are open dense subsets, then $\bigcap_{n\geq 1} U_n$ is dense.*

Proof. Let U_n be as in the statement of the lemma. After replacing U_n by $U_1 \cap \ldots \cap U_n$, we may assume that $U_1 \supset U_2 \supset \ldots$. Let M_n, $n \in \mathbf{N}$ be a fundamental system of neighbourhoods of 0. We may assume that $M_{n+1} \subset M_n$. Pick $x \in M$. We will show that for every $k \geq 1$ there exists a $y \in \bigcap_{n\geq 1} U_n$ with $x - y \in M_k$.

To construct y we argue as follows. First, we pick a $y_1 \in U_1$ with $y_1 \in x + M_k$. This is possible because U_1 is dense and $x + M_k$ is open. Then we pick a $k_1 > k$ such that $y_1 + M_{k_1} \subset U_1$. This is possible because U_1 is open. Next, we pick a $y_2 \in U_2$ with $y_2 \in y_1 + M_{k_1}$. This is possible because U_2 is dense and $y_2 + M_{k_1}$ is open. Then we pick a $k_2 > k_1$ such that $y_2 + M_{k_2} \subset U_2$. This is possible because U_2 is open.

Continuing in this fashion we get a converging sequence y_i of elements of M with limit y. By construction $x - y \in M_k$. Since

$$y - y_i = (y_{i+1} - y_i) + (y_{i+2} - y_{i+1}) + \ldots$$

is in M_{k_i} we see that $y \in y_i + M_{k_i} \subset U_i$ for all i as desired. □